

Aluminum 3003-O

Categories: [Metal](#); [Nonferrous Metal](#); [Aluminum Alloy](#); [3000 Series Aluminum Alloy](#)

Material Notes: Good weldability/formability and very good corrosion resistance lead to applications such as food and chemical handling, tanks, trim, litho sheet, pressure vessels, and piping.

Data points with the AA note have been provided by the Aluminum Association, Inc. and are NOT FOR DESIGN.

Composition Notes:

Composition information provided by the Aluminum Association and is not for design.

Key Words: UNS A93003; ISO AlMn1Cu; Aluminium 3003-O; AA3003-O; Al3003-O

Vendors: [Click here](#) to view all available suppliers for this material.

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Physical Properties	Metric	English	Comments
Density	2.73 g/cc	0.0986 lb/in ³	AA; Typical
Mechanical Properties	Metric	English	Comments
Hardness, Brinell	28	28	AA; Typical; 500 g load; 10 mm ball
Tensile Strength, Ultimate	110 MPa	16000 psi	AA; Typical
	18.0 MPa	2610 psi	
	@Temperature 400 °C	@Temperature 752 °F	
	29.0 MPa	4210 psi	
	@Temperature 300 °C	@Temperature 572 °F	
	60.0 MPa	8700 psi	
	@Temperature 200 °C	@Temperature 392 °F	
	90.0 MPa	13100 psi	
	@Temperature 100 °C	@Temperature 212 °F	
	110 MPa	16000 psi	
	@Temperature 25.0 °C	@Temperature 77.0 °F	
	115 MPa	16700 psi	
	@Temperature -30.0 °C	@Temperature -22.0 °F	
	150 MPa	21800 psi	
	@Temperature -100 °C	@Temperature -148 °F	
	230 MPa	33400 psi	
	@Temperature -200 °C	@Temperature -328 °F	
	97.0 - 130 MPa	14100 - 18900 psi	
	@Thickness 0.152 - 76.2 mm	@Thickness 0.00600 - 3.00 in	
Tensile Strength, Yield	41.4 MPa	6000 psi	AA; Typical
	12.0 MPa	1740 psi	
	@Temperature 400 °C	@Temperature 752 °F	
	17.0 MPa	2470 psi	
	@Temperature 300 °C	@Temperature 572 °F	
	30.0 MPa	4350 psi	
	@Temperature 200 °C	@Temperature 392 °F	
	38.0 MPa	5510 psi	
	@Temperature 100 °C	@Temperature 212 °F	
	41.0 MPa	5950 psi	
	@Temperature 25.0 °C	@Temperature 77.0 °F	
	45.0 MPa	6530 psi	
	@Temperature -30.0 °C	@Temperature -22.0 °F	
	52.0 MPa	7540 psi	
	@Temperature -100 °C	@Temperature -148 °F	
	60.0 MPa	8700 psi	
	@Temperature -200 °C	@Temperature -328 °F	
	>= 34.0 MPa	>= 4930 psi	
	@Thickness 0.152 - 76.2 mm	@Thickness 0.00600 - 3.00 in	
Elongation at Break	40 %	40 %	
	@Temperature 25.0 °C	@Temperature 77.0 °F	



	41 % @Temperature -30.0 °C	41 % @Temperature -22.0 °F	
	43 % @Temperature -100 °C	43 % @Temperature -148 °F	
	43 % @Temperature 100 °C	43 % @Temperature 212 °F	
	46 % @Temperature -200 °C	46 % @Temperature -328 °F	
	60 % @Temperature 200 °C	60 % @Temperature 392 °F	
	70 % @Temperature 300 °C	70 % @Temperature 572 °F	
	75 % @Temperature 400 °C	75 % @Temperature 752 °F	
	14 - 25 % @Thickness 0.152 - 76.2 mm	14 - 25 % @Thickness 0.00600 - 3.00 in	
	30 % @Thickness 1.59 mm	30 % @Thickness 0.0625 in	AA; Typical
	40 % @Diameter 12.7 mm	40 % @Diameter 0.500 in	AA; Typical
Modulus of Elasticity	68.9 GPa	10000 ksi	AA; Typical; Average of tension and compression. Compression modulus is about 2% greater than tensile modulus.
Notched Tensile Strength	110 MPa	16000 psi	2.5 cm width x 0.16 cm thick side-notched specimen, $K_t = 17$.
Ultimate Bearing Strength	207 MPa	30000 psi	Edge distance/pin diameter = 2.0
Bearing Yield Strength	83.0 MPa	12000 psi	Edge distance/pin diameter = 2.0
Poissons Ratio	0.33	0.33	
Fatigue Strength	48.3 MPa @# of Cycles 5.00e+8	7000 psi @# of Cycles 5.00e+8	completely reversed stress; RR Moore machine/specimen
Machinability	10 %	10 %	0-100 Scale of Aluminum Alloys
Shear Modulus	25.0 GPa	3630 ksi	
Shear Strength	75.8 MPa	11000 psi	AA; Typical

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.00000349 ohm-cm @Temperature 20.0 °C	0.00000349 ohm-cm @Temperature 68.0 °F	AA; Typical
Magnetic Susceptibility	8.0e-7	8.0e-7	cgs/g at 25°C

Thermal Properties	Metric	English	Comments
CTE, linear	21.5 $\mu\text{m/m-}^\circ\text{C}$ @Temperature -50.0 - 20.0 °C	11.9 $\mu\text{in/in-}^\circ\text{F}$ @Temperature -58.0 - 68.0 °F	
	23.2 $\mu\text{m/m-}^\circ\text{C}$ @Temperature 20.0 - 100 °C	12.9 $\mu\text{in/in-}^\circ\text{F}$ @Temperature 68.0 - 212 °F	AA; Typical; average over range
	24.1 $\mu\text{m/m-}^\circ\text{C}$ @Temperature 20.0 - 200 °C	13.4 $\mu\text{in/in-}^\circ\text{F}$ @Temperature 68.0 - 392 °F	
	25.1 $\mu\text{m/m-}^\circ\text{C}$ @Temperature 20.0 - 300 °C	13.9 $\mu\text{in/in-}^\circ\text{F}$ @Temperature 68.0 - 572 °F	
Specific Heat Capacity	0.893 J/g-°C	0.213 BTU/lb-°F	
Thermal Conductivity	193 W/m-K	1340 BTU-in/hr-ft ² -°F	AA; Typical at 77°F
Melting Point	643 - 654 °C	1190 - 1210 °F	AA; Typical range based on typical composition for wrought products $\geq 1/4$ in. thickness
Solidus	643 °C	1190 °F	AA; Typical
Liquidus	654 °C	1210 °F	AA; Typical

Processing Properties	Metric	English	Comments
Annealing Temperature	413 °C	775 °F	Commercial practice: 750 to 1100°F, higher temperatures used only for flash annealing.

Component Elements

Properties	Metric	English	Comments
Aluminum, Al	96.7 - 99 %	96.7 - 99 %	As remainder
Copper, Cu	0.05 - 0.20 %	0.05 - 0.20 %	
Iron, Fe	<= 0.70 %	<= 0.70 %	
Manganese, Mn	1.0 - 1.5 %	1.0 - 1.5 %	
Other, each	<= 0.05 %	<= 0.05 %	
Other, total	<= 0.15 %	<= 0.15 %	
Silicon, Si	<= 0.60 %	<= 0.60 %	
Zinc, Zn	<= 0.10 %	<= 0.10 %	

[References](#) for this datasheet.

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